

The Prose Barrier: Structural Limits of Agent Self-Verification and a Five-Layer Configuration Integrity Architecture

Yuhao Lin

Fujian Agriculture and Forestry University
3235706088@stu.fafu.edu.cn

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Abstract

AI coding agents are configured through natural-language rules—system prompts, markdown files, hook registries. Over long conversations, these rules degrade. The root cause, I find, is structural: an LLM’s generation and self-evaluation share the same decoder $P(\text{token} \mid \text{context}; \theta)$, so the model cannot independently verify rule compliance—it can only *generate a claim* that it did. I call this the Prose Barrier.

I present a five-layer architecture organized by position relative to the Barrier: L0 psychological safety (uncertainty admission), L1 mechanical gates (filesystem checks bypassing NL entirely), L2 neural gates (logprob probes detecting constraint penetration without content interpretation), L3 causal encoding (syllogistic format altering internal representation depth), and L4 drift prediction (trend-based early warning). Deployed over 34 agent sessions with 11 experiments across 3 architectures (DeepSeek V4 Pro, Qwen3-8B, GLM-4-9B).

Three patterns are visible. First, rules improve compliance (direction consistent across three architectures; e.g., +0.38 above baseline for DeepSeek), though behavioral scoring reliability was poor ($\kappa = -0.14$). Second, syllogistic format deepens constraint internalization at the logprob level ($d_z = +0.578$, $BF_{10} = 2.8 \times 10^5$, objective API-read DV) without changing behavioral output (IMP $\approx$ SYL across architectures). Third, format effects follow a non-monotonic pattern ($C_1 > C_0 > C_3 > C_2$) governed by output constraint severity, not a slope.

All experiments by a single undergraduate student on one laptop (Dell G15, RTX 3060). No supervisor. No funding. Code and data at <https://github.com/YuhaoLin2005/hermes-workspace>.

1 Introduction

A typical Claude Code agent loads 5–10 markdown files defining its behavioral rules: how to review code, when to run tests, what constitutes a deliverable. These rules sit in system prompts and hook registries. The agent reads them at session start. Then it works—sometimes for hours, across dozens of tasks, generating thousands of tokens.

Whether the agent actually follows its rules is an empirical question.

I ran 34 sessions with a rule-configured agent. Retrospective coding of these sessions showed rule violations in 55.9% of sessions (19 of 34) before mechanical enforcement was wired in. After adding filesystem-level checks (mtime comparison, regex matching, exit code verification) and running a 150-task controlled experiment across 6 sessions, the violation rate dropped to 0.7% (1 of 150 tasks). These behavioral numbers come from a scoring protocol later found to have poor inter-rater reliability ($\kappa = -0.14$; see Limitations), so they should be treated as directional indicators, not precise estimates.

The scoring protocol’s failure was itself informative: the same structural constraint that prevents the agent from self-verifying (Section 2) also prevented the single author from blind-scoring reliably—a mirror of the Prose Barrier at the meta-level, where independent verification requires a second person rather than a second prompt.

The core question is *why*. A language model that performs at professional-exam level can fail to reliably verify that it wrote a file to disk. The answer, I came to believe, is about architecture, not capability.

The core problem is that when an LLM agent evaluates itself—“Did I follow rule R on this task?”—its answer comes from the same decoder that may have already failed to follow R. There is no independent channel. The act of self-checking

and the act of generating share the same token distribution $P(\text{token} \mid \text{context}; \theta)$. This is what I call the **Prose Barrier**: a structural constraint that makes unassisted agent self-verification unreliable, not because the model is bad at it, but because the architecture makes genuine independent verification impossible.

I advance three claims:

1. **The Prose Barrier is a structural constraint, not an engineering bug.** It follows from the autoregressive nature of LLMs and means that verification channels must be architecturally independent of generation channels. This reframes agent reliability from “write better prompts” to “design architectures that work around the Barrier.”
2. **A five-layer architecture systematically addresses this constraint.** The layers are organized by their position relative to the Prose Barrier—pre-processing (L0), operating outside it (L1), probing inside it without interpreting content (L2), altering internal routing through it (L3), and observing it from outside across time (L4).
3. **Format effects on internal representations are measurable at the logprob level, but behavioral claims require independent validation.** Logprob-level measurement (L2, $d_z = +0.578$, $BF_{10} = 282k$) provides objective evidence that syllogistic rule format alters internal constraint representations compared to imperative format—without relying on human scoring. Constraint gradient experiments reveal a non-monotonic relationship between output constraint severity and format effect strength. The behavioral compliance numbers ($55.9\% \rightarrow 0.7\%$) are exploratory rather than confirmatory because all behavioral experiments were scored by a single unblinded rater; a partial blind check yielded $\kappa = -0.14$ —below chance. This limitation is discussed explicitly, and Section 4.4 specifies what would be needed for confirmatory claims.

These claims connect through a single mechanism: a rule that an agent follows is, at the representation level, a constraint that the decoder respects during generation. If format changes how deeply a constraint penetrates internal representations (Claim 3), then format engineering is a

lever for rule compliance. The logprob experiments below measure this penetration at the token-probability level, bypassing the scoring reliability problem that affects the behavioral data. The behavioral experiments test whether penetration translates to observable compliance. The gap between them—format changes representations but not behavior—is itself the central empirical finding.

2 The Prose Barrier

2.1 Definition

Consider an autoregressive language model M with parameters θ . At decoding step n , the model samples:

$$t_n \sim P(\cdot \mid t_1, \dots, t_{n-1}; \theta)$$

Every token the model produces—whether it is “thinking about what to do,” “evaluating what it just did,” or “taking action”—is drawn from this same distribution. There is no separate “auditor” module with independent access to ground truth. When the agent is asked “did you follow rule R ,” its answer is sampled from the same decoder whose state may already encode a failure to follow R .

Consequence. The Barrier cannot be circumvented by prompting. Telling the agent “be more careful about self-checking” generates more self-checking tokens—but those tokens are still drawn from the same distribution. The model cannot step outside its own generation to independently audit it. This is not a claim about how well LLMs self-verify; it is a claim about what self-verification *means* in an autoregressive architecture.

2.2 The Five-Layer Response

If the Barrier cannot be removed, it can be worked around. This configuration integrity architecture addresses agent rule compliance from five positions relative to the Barrier:

L0 — Psychological Safety. Pre-processes the generation by reframing uncertainty. Rather than “always be helpful and confident,” the agent is told: “Accuracy > completeness. ‘I don’t know’ is a valid output. You will not be punished for admitting uncertainty.” This does not bypass the Barrier—it changes what the model generates before the Barrier becomes relevant. Tested on 40 binary-choice probes: accuracy preserved (+0.01),

non-ceiling-probe uncertainty admission improved ($r = +0.949$, $n = 5$ probes below ceiling, 95% CI [0.57, 0.996]). **Status:** preliminary signal only; $n=5$ non-ceiling probes, a single outlier could drive the correlation.

L1 — Mechanical Gate. Operates entirely outside the Barrier. Instead of asking the agent “did you read back the file?” (NL, goes through the Barrier), a shell script compares mtime timestamps (filesystem, no NL involved). Violations are detected by regex matching, exit code checking, and hook wiring verification—none of which go through the model. Deployed over 34 sessions: violation rate dropped from 55.9% to 0.7% (exploratory; see Limitations for $\kappa = -0.14$ on behavioral scoring). 19 automated tests (e.g., hook-file mtime verification, SHA256 chain integrity, stale output detection via regex, exit-code gate enforcement) operate at the filesystem level, bypassing the model entirely; all 19 pass. **Status:** deployed and operational; single-machine only.

L2 — Neural Gate. Probes inside the Barrier but avoids content interpretation. Rather than reading the agent’s output and judging whether it “seems compliant” (an NL task, unreliable per the Barrier), I measure token logprobs at specific decision points. If a syllogistic rule format produces higher probability for rule-consistent choice tokens compared to imperative format, I have detected format penetration—without interpreting a single word of the agent’s output. The DV is read directly from the API JSON response (logprobs=True, top_logprobs=20); zero human scoring. **Status:** 40-probe validation complete ($d_z = +0.578$, $BF_{10} = 282k$); single model (DeepSeek V4 Pro) due to API logprob availability.

L3 — Causal Encoding. Alters internal processing through the Barrier. The hypothesis: reformulating the same rule from imperative (“verify your output”) to syllogistic form (“Write operations can deviate from intent. Only a read-back confirms correctness. Therefore, verify after writing.”) changes attention routing because the causal chain structure provides a template that the

Layer	Position	Mechanism	
L0 Safety	Pre-barrier	Reframe uncertainty admission	Accuracy $r = +0.949$
L1 Gate	Outside	Filesystem checks	55.9% → 0.7%
L2 Gate	Inside	Logprob probes	$d_z = +0.578$
L3 Encoding	Through	Syllogistic format	$d_z = 0.09$
L4 Predict	Outside	Drift features	12 features c

Table 1: Five-layer architecture: position relative to the Prose Barrier, mechanism, and current empirical support. L1 and L4 operate outside the Barrier (mechanical checks); L2 probes inside without interpreting content; L3 alters internal processing through the Barrier; L0 reframes before generation begins.

model’s attention mechanism can follow. Evidence from L2 logprob measurement, constraint gradient experiments, and convergent validation across two independent manipulations. **Status:** multiple experiments completed; causal mechanism relies on Pender (2026) for independent mechanistic evidence.

L4 — Drift Prediction. Observes L1–L3 state across time, entirely outside the Barrier. 12 mechanical features (SHA256 chain integrity, hook wiring consistency, stale file detection, etc.) are tracked across sessions. When feature patterns deviate from calibration baselines, the system flags emerging drift before behavioral failures cascade. **Status:** deployed with 12 features; predictive validation pending.

3 Key Experiments

3.1 Logprob V3: Does Format Change Internal Representation?

An initial attempt ($n = 8$ probes) produced $d_z = -0.148$ —a near-zero result in the wrong direction. I wrote a blog post about the null finding¹. A reader identified the issue: 4 of 8 probes had comparison tokens absent from DeepSeek’s top-20 logprobs, so my script substituted a -10.0 sentinel value—injecting noise that masked the real effect.

I built a mechanical pre-screening pipeline (probe_validator.py) that queries the API for top-20 logprobs at the decision token position and verifies both option tokens are present. Probes that

¹DEV.to: “My Experiment Showed Zero Effect. A Statistician Told Me My Measurement Was Broken.”

Category	IMP	SYL	d_z
Action	-3.66	+1.85	+0.50
Epistemic	-4.23	+3.92	+0.76
Structural	-2.37	+5.56	+0.67
Meta	-1.54	+3.21	+0.38

Table 2: Logprob V3 per-category results. All categories favor syllogistic format. Category $\times$ Format ANOVA: $F(3, 36) = 0.26$, $\eta^2 = 0.02$, n.s.

fail are rejected before the main experiment runs. This is the equivalent of calibrating an instrument before taking readings.

Design. Within-probe, 3-condition (NO RULES / imperative / syllogistic). 40 probes (10 per category: action, epistemic, structural, meta), all passing pre-screening. Each probe is a binary choice (A or B) embedded in a scenario drawn from real Claude Code agent usage. For instance: “*The agent completed a multi-file refactoring. Should it (A) run the test suite before declaring completion, or (B) trust the changes and proceed?*” —a probe for the action category testing whether the agent prioritizes verification over speed. The DV is $\log P(A) - \log P(B)$ at the first token position where both A and B logprobs are available, read directly from the DeepSeek V4 Pro API response ($T = 0.2$, chosen as a compromise: greedy $T = 0$ returns unavailable-logprob sentinels on this API, while $T = 1.0$ introduces excessive variance for a repeated-measures design; $T = 0.2$ provides usable logprobs with modest sampling noise).

Results. $n = 40$ probes. Mean IMP (imperative-format) differential: -2.95 ($SD = 14.67$). Mean SYL (syllogistic-format) differential: $+4.48$ ($SD = 13.45$). Paired difference $\bar{x}_\Delta = +7.43$, $SE = 2.03$, $t(39) = 3.65$, $p = 0.0008$ (two-tailed). Cohen’s $d_z = +0.578$ (bootstrap 95% CI on raw logprob differential: $[+3.39, +11.17]$). Bayes factor $BF_{10} = 282,399$ (JZS prior, $r = 0.707$), extreme evidence for the alternative. 32/40 probes (80%) show direction favoring syllogistic format.

The $SE = 2.03$ yields a 95% confidence interval of approximately $[+3.39, +11.17]$ on the raw logprob differential—the effect is robust but the magnitude is uncertain. Category $\times$ Format interaction was not significant ($F(3, 36) = 0.26$, $\eta^2 = 0.02$). The small η^2 indicates that the observed interaction effect is near zero ($\eta^2 = 0.02$);

however, with $n = 10$ per category, power to detect a medium interaction ($\eta^2 = 0.06$) at $\alpha = 0.05$ is approximately 0.35, so the absence of a detectable interaction should not be interpreted as evidence that no interaction exists—only that the observed effect size is small.

3.2 Where Format Helps: L1-Visibility Analysis

A reviewer of an earlier draft predicted that format effects should concentrate on rules where mechanical gates cannot reach—format as the “last line of defense” where L1 is absent. I tested the opposite prediction: format amplifies where L1 provides structural anchors.

Classification. Each of the 40 probes was classified as L1-visible (violation produces a deterministic, machine-detectable signal) or L1-invisible (violation concerns judgment quality, with no mechanical ground truth). The classification uses a three-test criterion: SIGNAL (does violation produce a deterministic mechanical signal?), ACTION (can a hook verify receipt-of-action?), CERTAINTY (zero false positives on mechanical signal?). All three must pass for L1-visible. 22/40 probes (55%) classified as L1-visible; 18/40 (45%) as L1-invisible.

Results. L1-visible probes: $\bar{x} = +9.47$, $s = 13.36$, within-group $d_z = +0.71$, 95% CI $[+0.35, +1.08]$. L1-invisible probes: $\bar{x} = +5.20$, $s = 12.91$, within-group $d_z = +0.40$, 95% CI $[+0.02, +0.78]$. The between-group difference $\Delta d = +0.31$ does not reach significance ($t(38) = 1.03$, $p = 0.31$) at $n = 40$ for a between-group comparison. A sensitivity analysis reclassifying three ambiguous probes at the boundary did not change the direction of the effect.

The prediction was wrong. Format helps *more* where L1 already operates (within-group $d_z = +0.71$ for L1-visible) than where it does not (within-group $d_z = +0.40$ for L1-invisible), though the between-group difference is not significant ($t(38) = 1.03$, $p = 0.31$). The direction is consistent with synergy (format amplifying existing structural anchors) rather than substitution (format compensating for missing mechanical enforcement), but $n = 40$ is underpowered for this between-group comparison and the classification was performed by a single unblinded rater.

Limitation. The classification was done by the same person who ran the experiment, without a second classifier or inter-rater reliability check.

The three-test criterion was developed to be as mechanical as possible (binary yes/no for each test), but the boundary between “receipt-of-action” and “receipt-of-diligence” has gray areas.

3.3 Constraint Gradient: A Non-Monotonic Pattern

The L2 neural gate measures format effect at the logprob level. But to measure it, I had to constrain the agent’s output—otherwise it produces paragraphs of natural language where the first-token logprob DV cannot be cleanly extracted. The constraint text was: “Output only the letter A or B” (Chinese original: “AB”).

This creates a paradox. I am measuring whether syllogistic format produces deeper reasoning about the constraint, but I am also telling the model to suppress its reasoning output. The measurement instrument is suppressing the mechanism it is measuring—what I came to call the **Prose Barrier of Measurement**.

This is not a paradox limited to the present design: any behavioral output constraint on an autoregressive model (forced-choice, short-form, structured format) risks suppressing the internal processing it aims to measure, because the same decoder generates both the reasoning and the response.

Design. To test whether this suppression is monotonic (stricter constraint $\rightarrow$ weaker format effect), I ran the 12 P1 probes through 4 levels of output constraint (labeled C0–C3 to distinguish from architectural layers L0–L4), 2 formats each, 96 API calls total (DeepSeek V4 Pro, $T = 0.2$):

- **C0:** No constraint. V3 baseline condition.
- **C1:** “Output only the letter A or B” (light: standardize output format).
- **C2:** “Output only one letter, no other text, no explanation” (moderate: explicitly suppress explanation).
- **C3:** “Do not output any characters except A or B. No explanation. No punctuation. No spaces. No newlines.” (heavy: maximal output suppression).

Results. The monotonic prediction is wrong.

The pattern (Table 3) suggests three qualitatively different processing regimes rather than a single monotonic function. **Important caveat:**

Level	n	$\bar{x}$	s	d_z
C0 (none)	11	+4.91	15.58	0.315
C1 (light)	12	+7.10	11.92	0.596
C2 (moderate)	12	+1.13	12.40	0.091
C3 (heavy)	12	+2.21	7.42	0.297

Table 3: Constraint gradient: format effect rises, crashes, then partially recovers. C0 $n = 11$ due to one probe (degradation-chain FATAL IMP) having no A/B token at the first position. The C0 $d_z = 0.315$ vs. the V3 $d_z = 0.578$ (Section 3.1) differ because the constraint gradient uses only the 12 P1 probes for within-experiment comparability, while V3 uses the full 40-probe pool. Constraint levels C0–C3 are distinct from architectural layers L0–L4.

no formal statistical test of non-monotonicity was conducted (e.g., quadratic contrast or piecewise regression across the four levels). The three-regime description below is a qualitative interpretation of the observed means, not a confirmed statistical model, and was developed on the same data used to evaluate it:

Optimization ($C_0 \rightarrow C_1$): d_z rises from 0.315 to 0.596. A light constraint that standardizes output format without telling the model to skip reasoning actually *improves* the measurement—clean parseable output, still-processed reasoning. This is the best measurement condition for logprob-based format effect detection.

Suppression ($C_1 \rightarrow C_2$): d_z crashes from 0.596 to 0.091. “No explanation” directly instructs the model to skip the reasoning channel that syllogistic format uses. The measurement constraint destroys the mechanism it is measuring. C2 is the worst measurement condition—it kills the mechanism while still requiring constrained output.

Rebound ($C_2 \rightarrow C_3$): d_z recovers to 0.297 with the tightest variance across all levels ($s = 7.42$). Under extreme compression, the model’s system prompt becomes the only remaining behavioral differentiator. Syllogistic format’s explicit causal structure may be embedding more effectively into compressed decision pathways than imperative brevity. But the constraint text is also longer (29 vs. 21 characters), and this length confound cannot be cleanly separated from severity.

Convergent evidence. The L1-visibility pattern transitions from synergy to compensation as constraint severity increases: Δ_{V-I} : $+10.98 \rightarrow +0.36 \rightarrow -2.03 \rightarrow -3.80$. The same reversal was observed in a separate experiment where multi-scene cognitive load was the manipulation ($\Delta = -10.08$ in the P1 experiment). Two independent manipulations producing the same qualitative pattern is suggestive—not conclusive, given the small n and post-hoc nature of the comparison, but warrants formal replication.

Per-probe variation. Two probes (self-audit-complexity and self-audit-logic) survive all four constraint levels with broadly stable format effects, while the shortest-chain L1-gatable probe (execution-rule-script) collapses entirely (f_x : $+16 \rightarrow +4 \rightarrow 0 \rightarrow -5$). Probes with moderate-complexity causal chains—not the simplest and not the most complex—show the most robust format effects. This is consistent with an inverted-U relationship between chain complexity and format benefit, though $n = 12$ is far too small to estimate a parametric curve.

3.4 Behavioral Replication Across Architectures

Rules improve compliance, with caveats. I ran the same 12 probes in a condition with all mechanical enforcement hooks removed (the agent’s rule-configuration system, called GateGuard, was set to OFF—all filesystem-level checks at L1 and behavioral-test hooks were disabled) with a NO RULES baseline on three models. Compliance rates (NO RULES $\rightarrow$ IMP $\rightarrow$ SYL): DeepSeek V4 Pro (MoE): $0.48 \rightarrow 0.86 \rightarrow 0.83$; Qwen3-8B (Dense): $0.79 \rightarrow 1.00 \rightarrow 0.98$; GLM-4-9B (GLM): $0.83 \rightarrow 1.00 \rightarrow 1.00$. These numbers are behavioral scores from a protocol with $\kappa = -0.14$ and should be interpreted as exploratory direction indicators, not precise estimates.

Format does not change behavior. IMP compliance $\approx$ SYL compliance on all three models ($|\Delta| \leq 0.03$). But on DeepSeek, Logprob V3 shows $d_z = +0.578$. Format changes internal representations without changing behavioral output—the same dissociation observed in the original GateGuard-OFF experiment now replicated on two additional architectures. That said, Qwen and GLM show near-ceiling compliance (IMP/SYL both at 0.98–1.00), so there is little room for format to create a behavioral difference even if one exists.

Cross-model constraint gradient. I ran the same 4-level constraint gradient experiment on Qwen3-8B and GLM-4-9B (behavioral measurement only; logprobs unavailable via SiliconFlow API). Neither model showed any format effect at any level (GLM: $d_z = 0$ everywhere, with perfect 12/12 A-selection in all conditions). Two possible explanations: (a) 8B–9B models genuinely lack the processing depth to benefit from syllogistic structure, or (b) the behavioral measurement method—which already shows $\text{IMP} \approx \text{SYL}$ on DeepSeek—cannot detect format effects at any model scale when ceiling effects are present. Logprob-level cross-model data would distinguish these explanations but was not available.

4 Discussion

4.1 What This Work Is and Is Not

This work has two contributions. The conceptual contribution is reframing agent reliability from “how to make them better” to “what makes self-verification structurally impossible”—identifying the Prose Barrier as the root cause of a practical engineering problem (agent configuration drift). The empirical contribution is demonstrating that format effects on internal constraint representations are measurable at the logprob level with an objective, API-read dependent variable—a method that bypasses the behavioral scoring problem ($\kappa = -0.14$, see Limitations) that prevents confirmatory behavioral claims in the current data.

This is not a completed project by the standards of a supervised lab; all behavioral results should be interpreted as exploratory (see Limitations for the full list).

4.2 Related Work

The Prose Barrier concept connects to several established results but also faces challenges from work demonstrating that LLMs can improve through internal processes. Wei et al. (2022) showed that chain-of-thought prompting elicits better reasoning; Wang et al. (2023) demonstrated that majority voting over multiple samples improves accuracy; and Madaan et al. (2023) found that iterative self-feedback can refine outputs. These results appear to challenge the Prose Barrier: if LLMs can self-improve, why can they not self-verify? The answer is that these methods all operate by changing the generation distribution $P(\text{token} \mid \text{context})$ —longer reasoning

chains, multiple samples, iterative refinement—rather than by providing an independent verification channel. They improve outputs by altering the distribution the decoder samples from rather than providing ground-truth access. This distinction matters: a model can get better at producing rule-consistent outputs through internal processes, but it still cannot independently verify that a specific output satisfies the constraint. External verification (L1) and internal improvement (CoT, self-consistency) are complementary, not substitutes. The Prose Barrier does not claim self-improvement is impossible; it claims that ground-truth-independent verification is structurally unavailable in a single-decoder architecture.

First, Kambhampati et al. (2024) established that LLMs require external verification modules—L1 is an instantiation of this principle for configuration management rather than planning, extending the LLM-Modulo framework to the problem of self-referential rule compliance. Bai et al. (2022) proposed constitutional AI where models self-critique their outputs; the Prose Barrier argues this self-critique is structurally constrained by the shared-decoder architecture, and that external verification channels (L1) are necessary wherever ground truth is mechanically checkable.

Second, Pender (2026) provided independent evidence that declarative vs. procedural format alters attention routing in LLMs; my L2/L3 experiments measure this phenomenon in an applied engineering context with logprob-level resolution. The non-monotonic constraint gradient extends this line of work by showing that the relationship between output constraint severity and format effect is not simply linear. Rath (2026) documented drift in multi-agent systems through inter-agent coordination metrics; this work focuses on single-agent configuration drift through the lens of a structural constraint (the Prose Barrier) rather than an emergent coordination phenomenon.

Third, Karpathy and Steinberger (2026) proposed deterministic external verification loops around probabilistic internal LLM loops—L1 is a configuration-domain instantiation of this principle. Czerwinski (2026) introduced the action-vs-diligence distinction that grounds the L1-visibility classification; the empirical finding that L1-synergy (amplification, not substitution) is the pattern adds quantitative evidence to this taxonomy.

Fourth, this work is shaped by two broader critiques of NLP research practice. Bender

and Koller (2020) argued that language models, trained on form alone, cannot in principle acquire meaning—a position that supports this caution against interpreting logprob shifts as evidence of “deeper understanding” rather than distributional preference changes. Lipton and Steinhardt (2018) identified patterns of overclaiming in ML scholarship, including failure to distinguish explanation from speculation and the use of suggestive terminology that carries unearned connotations; I have attempted to avoid these patterns through explicit hedging, pre-registration disclosure, and quantitative effect size reporting, though I acknowledge that terms like “Prose Barrier” and “Neural Gate” are novel coinages that carry rhetorical weight beyond their empirical grounding.

4.3 Practical Implications

Three implications follow from these findings for agent builders. These implications arise from a pattern that emerged across the full experimental sequence: the most reliable effects (L1 compliance gains, logprob format effects) are those measured through architecturally independent channels, while the least reliable (behavioral compliance scores) required a single unblinded rater. The reliability hierarchy mirrors the layers’ distance from the Prose Barrier.

First, rules that can be mechanically verified—where a script can detect a violation of rule R without NL interpretation—are fundamentally more enforceable than rules that require judgment. Second, the +0.38 compliance gain from having any rules (exploratory estimate) dwarfs format engineering effects in magnitude, suggesting that investment in L1 mechanical gates yields the highest marginal return. Third, if measuring format effects via logprobs, light output constraints preserve the reasoning channel being measured; heavy suppression destroys it.

4.4 What Would Be Needed for Confirmatory Claims

This paper reports exploratory findings from a single undergraduate student working without supervision, funding, or multi-model API access. Each of these requirements for confirmatory evidence is individually achievable: cross-model logprob replication costs an estimated \$2–5 in API calls, and the probe validation pipeline (probe_validator.py) is open-source at <https://github.com/YuhaoLin2005/>

[hermes-workspace](#). The barrier to confirmatory evidence is not resource but design—specifically, the need for independent blind scoring that the author cannot provide alone.

The following would be the minimum requirements to upgrade any of the behavioral claims to confirmatory status:

1. **Independent blind scoring.** At least two raters, blinded to condition, scoring all behavioral transcripts. Cohen’s $\kappa > 0.7$ is the conventional threshold for acceptable inter-rater reliability. The current $\kappa = -0.14$ (below chance) means the current scoring protocol cannot produce confirmatory evidence even in principle.
2. **Cross-model logprob replication.** Logprob V3 replicated on a non-MoE architecture (Claude, GPT-4) with logprobs API support, to rule out the possibility that the $d_z = +0.578$ format effect is specific to DeepSeek’s MoE architecture. Estimated cost: \$2–5 for 40-probe replication.
3. **Pre-registered constraint gradient hypothesis.** The three-regime model (optimization→suppression→rebound) should be pre-registered as a planned contrast with a formal test of non-monotonicity (e.g., quadratic contrast in repeated-measures ANOVA) on a new holdout probe set distinct from the $n = 12$ probes used for discovery.
4. **Second L1-visibility classifier.** The classification of probes as L1-visible vs. L1-invisible should be performed independently by a second rater, with classification agreement reported.

Until these conditions are met, all behavioral findings in this paper should be treated as hypothesis-generating rather than hypothesis-testing.

5 Limitations

The following limitations must be weighed when evaluating this work:

1. **Single-rater, unblinded behavioral scoring (fatal).** All behavioral experiments were scored by the author with knowledge of experimental conditions. A partial blind check

on $n = 8$ transcripts produced Cohen’s $\kappa = -0.14$ (95% bootstrap CI approximately $[-0.8, +0.4]$ with $n = 8$), indicating the scoring protocol does not meet the threshold for confirmatory inference. This low κ does not necessarily mean the raters disagreed on most cases—with near-ceiling compliance rates in the GateGuard-ON condition, extreme base rates can produce low kappa even when observed agreement is high (the “kappa paradox”; (Lipton and Steinhart, 2018), who warn against reporting metrics without context). Regardless of its cause, the $\kappa = -0.14$ means that all behavioral compliance numbers reported in this paper (including the 55.9% → 0.7% headline result) should be interpreted as exploratory, not confirmatory. Independent blind scoring by at least two raters achieving $\kappa > 0.7$ is the most critical precondition for any claim the current behavioral results support.

2. **Single-model logprob evidence.** Logprob V3 ($n = 40$) and the constraint gradient experiment ($n = 12 \times 4$ levels) were conducted on DeepSeek V4 Pro only. Cross-model family logprob replication (Claude, GPT-4) was prevented by API key and logprob availability constraints. The format effect findings may be specific to DeepSeek’s architecture (MoE with CSA/HCA).
3. **Small sample sizes in key analyses.** The constraint gradient uses $n = 12$ probes per level; confidence intervals are wide and the non-monotonic pattern could shift with additional probes. The three-regime model was discovered and described on the same data. L0 Safety Prompt correlation ($r = +0.949$) is based on $n = 5$ non-ceiling probes; a single outlier could drive this correlation. Cross-model behavioral replication uses $n = 12$ probes with $BF_{01} \approx 2.7$ —data favor the null but $n = 12$ is underpowered to distinguish a genuine null from an imprecise measurement.
4. **Probe pre-screening selection bias.** The `probe_validator.py` pipeline rejects probes whose option tokens do not appear in DeepSeek’s top-20 logprobs. This selects probes in which format effects are already large enough to push option tokens into the

measurable range, potentially inflating the pooled format effect estimate ($d_z = +0.578$). The direction and magnitude of this bias cannot be estimated from the current data.

5. **Constraint-length confound.** C2 and C3 constraint texts differ in length (21 vs. 29 Chinese characters). Severity cannot be fully isolated from prompt length in the constraint gradient experiment.
6. **Post-hoc L1-visibility classification.** The L1-visibility classification (§3.2) was performed by the author without a second classifier. While the three-test criterion (SIGNAL/ACTION/CERTAINTY) was designed to be mechanical, classification ambiguity at boundaries could affect the $d_z = +0.71$ vs. $+0.40$ comparison (within-group effects; the between-group difference $\Delta d = +0.31$ was not significant).
7. **Lack of pre-registration.** No experimental hypotheses or analysis plans were pre-registered. All reported analyses should be considered exploratory.
8. **Single author, single device, no supervision.** All experiments were conducted by a single undergraduate student on a single laptop (Dell G15, RTX 3060 6GB, 16GB RAM). No research supervisor reviewed the experimental design, data analysis, or manuscript. No funding supported this work.

6 Conclusion

Agent configuration integrity is not primarily a prompt engineering problem. It is an architectural problem with a structural root cause: the Prose Barrier. The five-layer architecture—permit (L0), bypass (L1), detect (L2), encode (L3), predict (L4)—offers one systematic response. Critically, only L1 constitutes an external verification channel (per Kambhampati et al. (2024)’s LLM-Modulo framework); L2 and L3 operate inside the model and measure internal representations, not independent verification. The empirical evidence has clear limitations (see Limitations above) but points in a consistent direction: rules work, format matters for internal processing but not behavioral output, and the effect of output constraints on format measurement is non-monotonic, forming three processing regimes rather than a sim-

ple slope. The Prose Barrier of Measurement—the discovery that the instrument used to measure format effects can suppress the mechanism being measured—has implications beyond this paper: it suggests that clean binary-output experimental designs for LLM evaluation may systematically underestimate format effects.

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The remaining errors, of which I am sure there are many, are mine alone.

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